

TECHNICAL & PRODUCT OVERVIEW

Gemin-Eye

AI-Powered Customer Acquisition Platform

How It Works

1. What is Gemin-Eye?

Gemin-Eye is an AI-powered customer acquisition platform that silently monitors online communities — Reddit, Facebook groups, LinkedIn feeds, Google Alerts, Quora, and more — looking for people who are actively asking questions that your business can answer.

When someone posts "Does anyone know a good bocce ball court in Chicago?" or "Looking for a graphic designer for my small business," Gemin-Eye catches it within minutes, scores it for purchase intent, crafts a natural, community-appropriate reply, and delivers the full package to the business owner's phone via Telegram — ready to copy and paste.

THE CORE IDEA

Instead of paying for advertisements that interrupt people, Gemin-Eye finds the moments when potential customers are already asking for help — and equips businesses to show up at exactly the right time with a genuinely helpful answer.

5 min

Reddit scan interval

2 min

Google Alerts interval

1–10

Intent score range

2–4

Sentence AI responses

Who Uses It

Gemin-Eye is a multi-tenant platform. A single installation serves multiple client businesses. Each client has one or more businesses, and each business has multiple campaigns targeting different channels. Business owners get their own lead feed; an admin can see and manage everything.

What Makes It Different

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2. System Architecture at a Glance

Gemin-Eye is a full-stack Node.js application with a React frontend, Express backend, and PostgreSQL database. It runs as a single deployable unit — no separate microservices — but is internally well-modularised.

Layer	Technology	Role
Frontend	React 18, TypeScript, Vite, TanStack Query, Tailwind	Dashboard, admin panel, onboarding wizard
Backend	Express.js, TypeScript, esbuild	REST API, background monitors, Telegram bot
AI	Google Gemini 2.5 Pro & Flash via @google/generative-ai	Scoring and response generation
Database	PostgreSQL + Drizzle ORM	All persistent state
Auth	Replit OIDC / Passport.js	Session-based authentication
Messaging	Telegram Bot API + Slack Incoming Webhooks	Lead delivery to clients
Monitoring	Custom RSS polling (Node.js setInterval)	Reddit & Google Alerts scanning

High-Level Data Flow

Internet (Reddit / Google Alerts / FB / LinkedIn) → Monitor / Bookmarklet / Telegram input → Keyword Matching → Gemini Flash Lead Scoring → Gemini Pro Response Generation → Telegram / Slack Delivery → Business Owner Feedback Loop

Key Design Decisions

Single deployment unit: The same Node process serves the REST API and the React frontend.

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5. Lead Discovery — How Gemin-Eye Finds Prospects

Gemin-Eye has four distinct channels for ingesting potential leads. Two are fully automated (running 24/7), one is semi-automated (user-triggered via bookmarklet), and one allows direct manual input.

3.1 Automated Reddit Monitor

The Reddit monitor (`server/reddit-monitor.ts`) runs every 5 minutes using Node's `setInterval`. On each cycle it:

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Queries the data base for all active Reddit campaigns to get their configuration and subscribed reddit ts.

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Fetches the RSS feed for each subreddit: `http`

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WHY RSS?

Reddit's RSS feeds are public and require no API key, avoiding OAuth complexity and rate limit tiers. Each subreddit has a /new.rss endpoint that returns the 25 most recent posts in standard XML.

3.2 Automated Google Alerts Monitor

The Google Alerts monitor (server/google-alerts-monitor.ts) runs every 2 minutes. Google Alerts can be configured to publish results as an RSS feed — Gemin-Eye polls these feeds and processes the results. The monitor parses the source URL to identify the originating platform (Quora, Reddit, Stack Overflow, news, forum) and applies platform-specific processing before scoring.

GOOGLE ALERTS SETUP

Google Alerts RSS feeds are created manually at google.com/alerts using target keywords. The owner pastes the RSS URL into their Gemin-Eye campaign. Gemin-Eye automatically creates a Google Alerts campaign for every new business during onboarding.

3.3 Spy Glass Bookmarklets (Facebook & LinkedIn)

The Spy Glass is a browser bookmarklet — a small JavaScript snippet saved as a browser bookmark. When clicked on a Facebook group or LinkedIn feed, it:

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3.4 Telegram Bot Input

Business owners can also paste content directly into the Telegram bot — raw post text, a URL, or even a screenshot. Gemini's vision capabilities perform OCR on images to extract text, which is then scored as normal.

4. AI Engine — Scoring & Response Generation

Every candidate post passes through a two-stage AI pipeline. Two different Gemini models are used intentionally — Flash is faster and cheaper for high-volume scoring; Pro produces higher quality, more nuanced prose for response generation.

4.1 Lead Scoring (Gemini Flash)

Once a post passes keyword matching, it is sent to Gemini 2.5 Flash for lead scoring. The prompt provides the model with the full post text, business profile (name, type, core offering, target audience), campaign keywords, and source platform. The model returns:

```
{ "score": 8, "reasoning": "Post author actively looking for bocce ball venue...", "isLead": true }
```

Score thresholds:

Channel	Minimum Score to Qualify
Automated monitors (Reddit, Google Alerts)	≥ 5
Manual scan (bookmarklet, Telegram)	≥ 4
Slack delivery	≥ 7

4.2 Response Crafting (Gemini Pro)

Qualified leads are passed to Gemini 2.5 Pro to generate a suggested reply. The model receives everything Gemini Flash received, plus feedback history from the owner and platform-specific tone instructions. Responses are always 2–4 sentences.

Platform	Response Style
Reddit	Sound like a regular user. Never mention the business by name. Never include links.
Facebook Groups	Warm, community-oriented. Business can be mentioned naturally.
LinkedIn	Professional tone. Can reference expertise and credibility.
Google Alerts (Quora/Forums)	Authoritative, long-form friendly. Position as an expert.

4.3 Keyword Matching Algorithm

Before any AI is invoked, a fast keyword filter narrows down which posts are worth scoring — saving API calls on clearly irrelevant posts. Key features:

- Must include keywords

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4.4 Mental Health Safety Layer

When post content contains mental health crisis signals (words like "psychosis", "schizophrenia", "suicidal", "crisis"), a dedicated safety module activates:

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SAFETY FIRST

This layer is non-negotiable and cannot be disabled by any campaign or admin configuration.
Human wellbeing always takes priority over lead generation.

5. Lead Delivery — Telegram & Slack

Telegram Notifications

Telegram is the primary delivery channel. Each qualified lead triggers a structured message sent to the business owner's configured chat ID. A typical lead message includes:

- Business name and campaign tag source (e.g. Reddit · r/AskChica go)
- I nte nt Sc ore (e.g. 8/10) wit h AI rea so nin g

- A snippet of the original post

- The AI-generated suggestions described responsible in a code block — a single tap copies it on mobile

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The response is formatted as a Telegram code block, making it a single tap to copy on mobile. The owner can paste it directly into Reddit without any editing if satisfied.

Slack Integration

Campaigns can optionally be configured with a Slack Incoming Webhook URL. When a lead scores 7 or above, Gemin-Eye also posts to the configured Slack channel using rich Block Kit formatting — same information as Telegram but structured for Slack's UI.

Leads with scores below 7 are still stored in the database and delivered via Telegram, but do not warrant a Slack ping. A business can have different Slack webhooks for different campaigns, routing different lead types to different Slack channels.

6. Feedback Loop — Teaching the AI

One of Gemin-Eye's most powerful features is its ability to improve over time based on owner feedback. This is a direct, interpretable feedback injection system — not vague machine learning.

How Feedback Works

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Button	Meaning	Effect on Future Responses
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Ø=ÜN Bad Match	Lead was not relevant to the business	Signals keyword matching may be too broad
Ø=Üâ Too Salesy	Response felt too promotional	AI instructed to reduce promotion and be more helpful

WHY THIS MATTERS

Every business owner has a unique voice and tolerance for promotion. The feedback loop means Gemin-Eye personalises itself to each business over time without any manual reconfiguration. The longer a business uses the platform, the better the responses get.

7. Client Onboarding

New businesses are onboarded through a guided multi-step wizard, available both on the web and through the Telegram bot. The process is designed to take less than 5 minutes.

Step 1 — Business Information

The owner provides their business website URL. Gemin-Eye scrapes the website to understand what the business actually does — extracting the business type, core offering, and target audience from the real site content rather than relying on self-description. This ensures the AI is always working with accurate context.

Step 2 — AI Strategy Generation

Gemini Pro analyses the scraped business data and generates a comprehensive monitoring strategy: target keywords and phrases, recommended subreddits, recommended Facebook group types, sample responses demonstrating the recommended tone, and notes on what to watch out for.

Step 3 — Telegram Connection

The owner connects their Telegram account so leads can be delivered. They are guided to start a conversation with the Gemin-Eye bot, which registers their chat ID automatically.

Automatic Campaign Creation

Upon completing onboarding, Gemin-Eye automatically creates: a Reddit campaign with AI-recommended subreddits, a Google Alerts campaign, and a Facebook campaign pre-configured for Spy Glass bookmarklet use.

8. Admin Panel

The admin panel is accessible only to super-admin accounts and provides a birds-eye view of the entire platform.

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9. Data Model & Storage

All data is stored in PostgreSQL via Drizzle ORM. The schema is defined in `shared/schema.ts` and shared between frontend and backend for full type safety.

Table	Description	Key Fields
users	Platform accounts	id, email, role, telegramChatId
businesses	Client businesses	id, userId, name, type, coreOffering, targetAudience, website
campaigns	Monitoring campaigns	id, businessId, type, keywords, subreddits, slackWebhookUrl, isActive
leads	Discovered prospects	id, campaignId, postId, postTitle, intentScore, source, isQualified
ai_responses	AI-generated replies	id, leadId, responseText, model, reasoning
response_feedback	Owner feedback	id, aiResponseId, feedbackType, createdAt
sessions	Express sessions	sid, sess, expire

All database operations go through typed TypeScript interfaces. No route handler touches the database directly — all calls go through the storage layer, ensuring consistent error handling and a clean boundary between business logic and persistence.

10. Security & Robustness

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API & Secrets Security



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Technology Stack Summary

Category	Technology	Notes
Language	TypeScript	Full-stack, shared types frontend & backend
Frontend Framework	React 18	Functional components, hooks throughout
Frontend Build	Vite	Dev server + production bundler
Frontend Routing	Wouter	Lightweight client-side router
State Management	TanStack React Query v5	Server-state synchronisation
UI Components	shadcn/ui + Radix UI	Accessible, composable primitives
Styling	Tailwind CSS	Utility-first; indigo/violet brand palette
Forms	React Hook Form + Zod	Controlled, validated forms
Backend Framework	Express.js	REST API + static file serving
Backend Build	esbuild	Fast TypeScript !' CJS compilation
Database	PostgreSQL	Primary data store
ORM	Drizzle ORM	Type-safe queries; drizzle-zod for validation
AI — Scoring	Google Gemini 2.5 Flash	High-speed intent scoring
AI — Responses	Google Gemini 2.5 Pro	High-quality response generation
Authentication	Replit OIDC + Passport.js	OpenID Connect, no passwords stored
Session Storage	connect-pg-simple	PostgreSQL-backed sessions
Lead Delivery	Telegram Bot API	Primary notification channel
Team Notifications	Slack Incoming Webhooks	Block Kit formatted messages
Deployment	Replit Autoscale	Always-on, auto-scaling hosting
Domain	gemin-eye.com	Production domain

OPEN-SOURCE FOUNDATIONS

Gemin-Eye is built entirely on open-source technologies with no proprietary third-party SaaS dependencies beyond Gemini AI, Telegram, and Slack — all of which have free tiers suitable for early-stage usage.

